



**Cal Poly
Pomona**

**College of Business
Administration**

Customer Satisfaction of Students Eating at Cal Poly Pomona Restaurants

MSDM, Cal Poly Pomona

IBM 6500: Customer Insights Methods & Survey Research

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Executive Summary

The purpose of our research was to cover satisfaction from on-campus dining experiences. We wanted to capture not only Cal Poly Pomona students as well as Cal Poly Pomona staff members to see how they respond. Our research consisted of primary data gathered from our survey as well as finding some background research to help us develop independent and dependent variables to help us with establishing our four hypotheses. The survey was established in October, and we ran the survey from the beginning of November to the end of the same month. The survey was distributed through physically by handing out or finding it on one of the discussion boards found in the business department and digital flyers distributed multiple days through Instagram. The physical flyer provided a QR code for survey takers to scan while the digital uses the link to take you to the site. We gathered 68 total responses consisting of mainly Cal Poly Pomona students and faculty members. We ran the data analysis based on the following: descriptive analysis, clustering, regression analysis and chi-squared tests. Based on our findings and running the data analysis, we discovered two out of four of our hypotheses proved to be supported and the other two vice versa. While we did not have a main client for on-campus dining experience, the overall is to look and see how students currently feel and if we were to share our findings with Cal Poly Pomona, we could look at how to improve overall satisfaction for everyone that attends Cal Poly Pomona.

Introduction

Understanding what drives student satisfaction with campus dining services represents more than an academic exercise. It connects directly to student wellbeing, campus retention, and the overall college experience. At Cal Poly Pomona, students interact daily with Foundation-operated dining venues including Centerpointe, eateries at the Bronco Student Center, and

various campus cafés. These interactions shape not only their immediate dining choices but also their broader perception of campus life and their sense of belonging within the university community.

This research investigates the factors that influence student satisfaction with Cal Poly Pomona's dining services, with particular attention to operational elements that often receive less scrutiny than food quality and pricing concerns. While much attention in campus dining services focuses on menu variety and cost considerations, students frequently express frustration with operational frictions such as excessive wait times, cleanliness, food quality, and problematic mobile ordering systems. These operational challenges may prove more influential in shaping dining behavior than previously recognized, particularly given the somewhat captive nature of the campus dining experience.

Background Analysis

The Cal Poly Pomona campus presents a unique dining context that shapes student behavior in distinctive ways. Unlike urban campuses where students can easily access numerous off-campus dining alternatives, Cal Poly Pomona's location creates a more contained dining environment. Students who spend extended hours on campus for classes, study sessions, and extracurricular activities must navigate the available Foundation dining options or develop alternative strategies for managing their meals.

This captive audience dynamic introduces an interesting complexity into satisfaction research. Traditional restaurant and food service research assume customers have genuine choices and can easily substitute for competitors when dissatisfied. However, campus dining operates under different constraints. Students may continue using on-campus dining despite dissatisfaction simply because alternatives require more time and effort than their schedules

permit. This raises important questions about how satisfaction translates into behavior in campus settings and what students do when they find campus options less than satisfactory.

When faced with disappointing campus dining experiences, students develop various coping strategies. Some bring meals prepared at home, investing time in meal preparation to avoid campus options. Others skip meals entirely while on campus, choosing to eat before arriving or after leaving rather than utilizing available services. Still others resort to extensive snacking rather than proper meals, purchasing items from vending machines or convenience stores. Understanding these alternative behaviors becomes crucial because they represent the true competitive set for campus dining services. The competition is not other restaurants but rather the option to opt out of formal dining altogether.

The Foundation dining services at Cal Poly Pomona must navigate this complex landscape while addressing multiple stakeholder needs. They must maintain financial viability, meet nutritional and health standards, accommodate diverse dietary requirements, and create positive experiences that support student success. Operational efficiency matters not only for the dining services' bottom line but also for student satisfaction, as efficient operations translate directly into shorter wait times, better food temperature, and reduced stress during already hectic academic schedules.

Literature Review

To understand campus dining satisfaction, we need to look at some established theories about how people make decisions and judge service quality. Two frameworks particularly help explain what's happening with our dining choices.

- **The Theory of Planned Behaviour**, developed by researcher Icek Ajzen (1991), explains that our behavior comes from three things: our attitudes (how we feel about something), subjective norms (what our friends and peers think), and perceived behavioral control (how easy or hard it is to actually do something). Think about deciding where to eat lunch. You might like the food at Centerpointe (attitude), your friends might always eat there (subjective norms), but if the line is always insanely long and you only have 20 minutes (low behavioral control), you might skip it anyway. That last factor, behavioral control, is where all those operational issues like wait times and hours come in. Research applying TPB to food choices has consistently shown that perceived behavioral control significantly influences whether people actually follow through on their eating intentions (McDermott, Oliver, Svenson et al., 2015).
- **The SERVQUAL framework** helps us think about service quality in dining. Originally developed by Parasuraman, Zeithaml, and Berry (1988), it identifies five dimensions that matter: tangibles (the physical environment and cleanliness), reliability (getting your order right consistently), responsiveness (speed of service), assurance (staff courtesy), and empathy (personalized attention). Research has successfully adapted this framework to restaurants and campus dining contexts. The five dimensions are often remembered using the acronym RATER: Reliability, Assurance, Tangibles, Empathy, and Responsiveness.

Studies specifically on college dining reveal some interesting patterns. Kim, Ng, and Kim (2009) studied university dining facilities using an adapted version of SERVQUAL called DINESERV. Their findings showed that food quality was the most important predictor of satisfaction, but operational factors like convenience, speed of service, and atmosphere also significantly influenced whether students were satisfied and whether they'd return. This research demonstrated that you can't just focus on food quality alone—the whole experience matters.

Ali and Ryu (2015) examined student foodservice experiences and found that overall experience significantly affects not just satisfaction but also how much students spend and how often they dine out. This is important because it shows satisfaction actually translates into behavior, not just positive feelings. Students who had better experiences came back more often and spent more money per visit.

More recent research by Lugosi (2019) took a broader view, examining how campus foodservice connects to student wellbeing. This integrative review noted that most existing research focused narrowly on price, food quality, and satisfaction with service, but didn't examine how operational factors and the dining environment affect students' broader college experience and health. Lugosi emphasized that campus dining spaces function similarly to co-working spaces, where the physical environment, seating arrangements, and atmosphere all contribute to social interaction and student wellbeing beyond just eating.

Research on mobile ordering technology in food service shows mixed results. Okumus and Bilgihan (2014) proposed that when mobile ordering works well, it should enhance customer satisfaction by reducing perceived wait times and increasing convenience. They found that factors like ease of use, usefulness, and enjoyment of the technology all influenced whether students intended to use mobile ordering systems in restaurants. However, more recent research

suggests the reality is more complex. Dirsehan and Cankat (2021) studied mobile food ordering apps during the pandemic and found that satisfaction with the mobile app experience itself was critical for developing overall restaurant satisfaction and loyalty. When the apps work well, they enhance the experience. But when they're buggy or create new problems, they can actually make things worse.

Xu, Liu, Mao, and Zhou (2024) conducted a large-scale study using real transaction data from Chinese restaurants and found something surprising: while mobile food ordering apps increased restaurant revenue through higher table turnover, they were actually negatively correlated with customer satisfaction in dine-in settings. The problem? When restaurants get overwhelmed with mobile orders during busy times, it can lead to longer wait times for both mobile and walk-in customers, creating frustration rather than convenience.

Research on campus dining specifically notes a gap in the literature. While numerous studies examine menu preferences and dietary needs, fewer quantify how operational frictions like wait times, limited hours, and seating availability compare against food quality concerns in driving student behavior. Most campus dining research has been quantitative, focusing on satisfaction ratings rather than examining the deeper links between dining experiences, operational efficiency, and student wellbeing (Lugosi, 2019). Our research addresses this gap by directly comparing operational versus food factors and measuring their relative impact on both satisfaction and actual dining frequency.

Problem Definition

Variables and Their Operational Definitions

- **Dependent Variable**

Overall satisfaction is our main outcome we're trying to explain. We'll measure it using several questions asking you to rate agreement with statements like "I'm satisfied with my overall experience at campus dining venues" and "Campus dining meets my needs." These use a five-point scale from Strongly Disagree to Strongly Agree.

We're also looking at **dining frequency** as a behavioral measure. This captures how often you use campus dining each week. Satisfaction is an attitude, but frequency shows whether that attitude translates into actual behavior.

- **Independent Variables**

- **Wait time** measures how long you typically spend from joining a line or ordering until you get your food. We'll ask you to characterize typical wait times and indicate whether they influence where you eat.
- **Hours of operation** captures whether dining venues are open when you need them. You'll tell us how often you find places closed when you want to eat and whether limited hours cause you to skip meals or find alternatives.
- **Seating availability** measures whether you can find a place to sit. We'll ask how often you struggle to find seating and whether this makes you avoid dining on campus during busy times.
- **Mobile ordering** functionality measures your experience with ordering apps like Grubhub. If you use mobile ordering, we'll ask about satisfaction with the process, order accuracy, and whether it actually saves time. If you don't use it, we'll ask why not.

- **Cleanliness** covers dining areas, food prep areas, and restrooms. We'll ask you to rate cleanliness and indicate whether cleanliness concerns affect your dining choices.
- **Food quality** includes taste, freshness, temperature, and portion size. We'll measure each separately to identify which specific aspects matter most to you.
- **Price** captures whether you think campus dining represents good value and whether cost affects how often you eat on campus.
- **Psychological Factors**
 - **Health consciousness** measures how much you prioritize nutritious choices.
 - **Variety seeking** captures whether you want diverse menu options or prefer consistent favorites.
 - **Time consciousness** measures how much you value speed and efficiency.
- **Lifestyle Factors**
 - **Commuter status** matters because living on or near campus versus commuting creates different dining patterns.
 - **Class schedule density** affects whether you have time for sit-down meals or need grab-and-go options.
 - **Campus activity involvement** may increase dependence on campus dining.
- **Demographics**
 - We'll collect standard information including academic year, major, and whether you have a meal plan. These help us identify whether different student groups experience campus dining differently.

Hypothesis:

- **H1 (Operations vs. Food/Price):** Operational issues (wait times, hours, seating, mobile pickup) have a stronger negative effect on on-campus dining choice than dissatisfaction with food quality or price.
- **H2 (Food Quality):** Higher perceived food quality (taste, freshness, temperature) is positively associated with overall satisfaction.
- **H3 (Mobile Ordering):** Students who frequently use mobile ordering systems (e.g., Grubhub) report higher satisfaction compared to non-users.
- **H4 (Cleanliness):** Perceived cleanliness/hygiene is positively associated with overall satisfaction.

Methodology

Our study used a quantitative, cross-sectional survey design to examine factors influencing student satisfaction with on-campus dining at Cal Poly Pomona. The target population was current CPP students who had eaten at least once at a campus dining venue (e.g., Centerpointe, BSC eateries, campus cafés). Data was collected using a self-administered online questionnaire hosted on Qualtrics.

The survey was developed in late October and fielded from early to late November. A non-probability convenience sampling strategy was used. The survey link was distributed through physical flyers containing a QR code placed around campus and via social media (Instagram) posts on two separate days. In total, 68 valid responses were collected.

The questionnaire contained 17 items, including multiple-choice questions, matrix-table Likert scales, Net Promoter Score, and open-ended text questions. Items measured background

characteristics (enrollment status, days and hours on campus, living status), dining frequency, primary venue, and perceptions of key attributes such as food quality, cleanliness, price fairness, wait times, hours of operation, seating availability, and mobile ordering. Overall satisfaction and likelihood to recommend were captured using agreement scales.

Data from Qualtrics were cleaned and coded before analysis. We first ran descriptive statistics to summarize dining frequency, satisfaction levels, and attribute ratings. We then used chi-square tests to explore relationships between categorical variables (e.g., commuter status and dining frequency), and regression analyses to test our hypotheses about the impact of operations, price, food quality, mobile ordering, and cleanliness on overall satisfaction. Finally, we conducted a cluster analysis to identify distinct student groups based on their dining behaviors and satisfaction profiles.

Survey Design

The final survey had an overall of 17 questions. This included the consent question, an even number of questions focusing on the independent variables we wanted to focus on (i.e. cleanliness, mobile ordering, quality, etc.). The total survey questions finalized at 17 questions. We had a mixture of multiple choice, selecting one response or selecting multiple responses available. For the form field responses this focused on students to put in numerical answers instead of doing a dropdown to where students could possibly make a mistake and to eliminate the need to scroll and look for their response to the question. The third type of question we had was the matrix table (Likert Scale). This question design proved to be helpful for our independent variables and ranking from a 1 to 5. The final question that proved to be helpful for our overall satisfaction for how the responder felt about their primary location and to see if they would recommend it to another classmate or friend was the Net Promoter Score. This question

type allowed responses to range from 1 to 10 to see how they felt overall satisfaction and if it was worth recommending. The text entry questions were considered optional and were not primary for our survey. The purpose of the text entry questions was to enter for the raffle or to see if our responders had anything else they wanted to share regarding the survey.

As far as having an initial survey and a finalized survey there were a few changes and additional questions that we implemented. One question that was implemented was the additional information question. We noticed from early responses that we had a response where the responder was adding additional information that they wanted to share, with that question being designated for the raffle only. With the addition of the question and making it optional it does allow us to look for possible future qualitative research. The changes focused on changing the question type. We changed the overall satisfaction and recommended the primary venue questions from a Likert Scale to the Net Promoter. This was due to the discovery of the question type and felt it looked better compared to a Likert Scale that focuses on having multiple questions tied to the main idea. Whereas if we kept it as a Likert Scale it was only 1 question, same with recommending the primary venue.

Data Analysis

All data analysis for this project was conducted using a combination of Python (Jupyter Notebook) and Power BI. After exporting the survey responses from Qualtrics into Excel, the dataset was imported into Python for cleaning, transformation, and statistical testing.

Data Preparation

We began by screening the dataset for missing or incomplete responses. Because the questionnaire used a 1–5 scale where 1 = strongly agree and 5 = strongly disagree, all perceptual items were reverse-coded to ensure that higher values represented more positive evaluations

(e.g., better operations, higher food quality, greater cleanliness, and higher satisfaction).

In Python, this was achieved using vectorized transformations in pandas.

Regression Analyses

To test the study's hypotheses, we performed several regression analyses in Python using the statsmodels library:

- **Multiple Regression (H1):**

We created an Operations composite score (hours, wait times, mobile ordering) and compared its impact on satisfaction against price fairness. This allowed us to assess which factor was more influential in predicting overall satisfaction.

- **Simple Linear Regressions (H2–H4):**

Individual regressions were run to test whether:

- Food quality (H2),
- Mobile ordering reliability (H3), and
- Cleanliness (H4)

each significantly predicted overall satisfaction.

Regression outputs (β coefficients, significance levels, R^2 values) were generated directly in Jupyter Notebook.

Segmentation Analysis

To better understand differences between groups of students, we also conducted a segmentation analysis in Python. Students were grouped based on their ratings of key service dimensions (operations, food, cleanliness). This allowed us to identify meaningful clusters—such as highly satisfied users, dissatisfied users, or operationally sensitive users—and compare their characteristics.

Chi-Square Test

To complement the regression analyses, we performed a chi-square test of independence using categorized variables. Specifically, we tested whether satisfaction levels differed across groups defined by how many days students were on campus per week. The chi-square statistic and contingency table were generated in Python.

Visualization and Mapping (Power BI)

In addition to Python plots, Power BI was used to create an interactive map and bar visualization showing visit frequency by dining venue. This helped identify which locations had the highest student traffic and supported interpretation of the regression findings.

Regression Analysis

Regression analyses were conducted in Python using Jupyter Notebook to evaluate the relationships proposed in the study's hypotheses. Prior to running the models, all Likert-scale items were reverse-coded so that higher values reflected more positive perceptions of operations, food quality, mobile ordering reliability, cleanliness, and overall satisfaction. This ensured consistency in interpretation across all analyses.

A multiple regression model was used to test H1, which examined whether operational factors (opening hours, wait times, and mobile ordering reliability) or price fairness had a stronger influence on overall satisfaction. An Operations composite score was created by averaging the three operational items, and both the composite and the price variable were entered as predictors in the model.

For H2, H3, and H4, simple linear regressions were conducted. Each model tested whether a single predictor—food quality (H2), mobile ordering reliability (H3), or cleanliness

(H4)—significantly predicted overall satisfaction. These models allowed us to isolate the effect of each service dimension and determine its unique contribution to student satisfaction.

All regression outputs, including coefficients, p-values, and R^2 values, were generated using the statsmodels package in Python. This approach provided a clear statistical framework for evaluating the hypotheses and determining which factors most strongly contributed to students' satisfaction with on-campus dining venues.

Hypothesis 1 Analysis

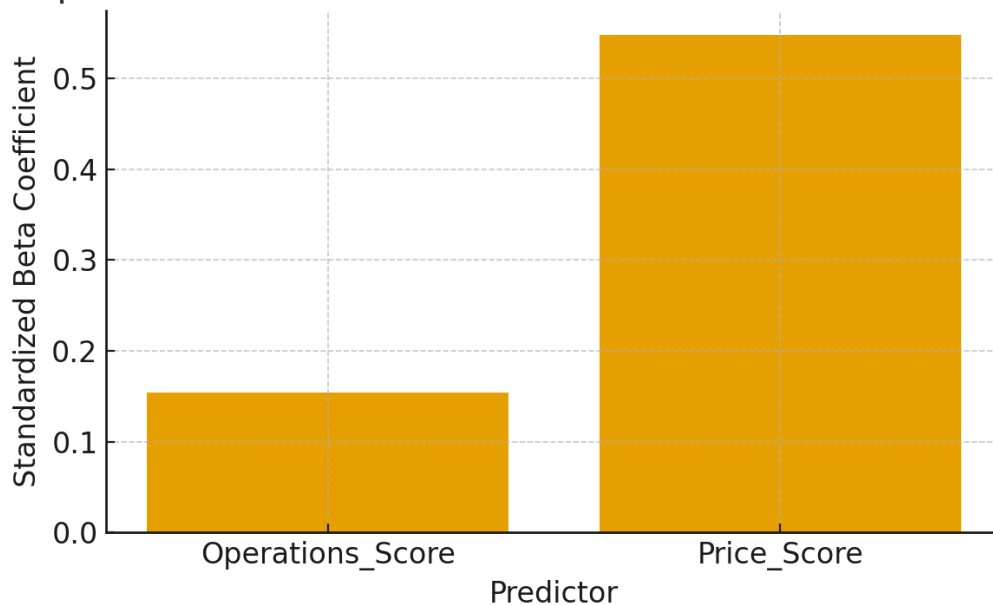
To test H1, a multiple regression analysis was conducted in Python using Jupyter Notebook. This hypothesis proposed that *operational factors*—including opening hours, wait times, and mobile ordering reliability—would have a stronger influence on overall satisfaction than *price fairness*. Prior to analysis, all survey items were reverse-coded so that higher scores reflected more positive evaluations.

An Operations composite score was created by averaging the three operational items (Q1_1, Q1_2, Q1_3). This composite and the price fairness item (Q1_4) were entered simultaneously into a multiple regression model, with overall satisfaction (Q3_1) serving as the dependent variable. The model allowed us to compare the relative predictive strength of operational performance versus price perceptions.

This analysis directly evaluated whether operational aspects of the dining experience outweighed price considerations in driving student satisfaction, providing a rigorous statistical test of H1.

OLS Regression Results							
Dep. Variable:		DV		R-squared:		0.347	
Model:		OLS		Adj. R-squared:		0.317	
Method:		Least Squares		F-statistic:		11.67	
Date:		Tue, 02 Dec 2025		Prob (F-statistic):		8.59e-05	
Time:		21:16:34		Log-Likelihood:		-44.483	
No. Observations:		47		AIC:		94.97	
Df Residuals:		44		BIC:		100.5	
Df Model:		2					
Covariance Type:		nonrobust					
		coef	std err	t	P> t	[0.025	0.975]
	const	2.8431	0.282	10.095	0.000	2.276	3.411
Operations_Score		0.0683	0.055	1.250	0.218	-0.042	0.178
Price_Score		0.2646	0.059	4.453	0.000	0.145	0.384
Omnibus:		3.181	Durbin-Watson:		2.332		
Prob(Omnibus):		0.204	Jarque-Bera (JB):		2.463		
Skew:		-0.557	Prob(JB):		0.292		
Kurtosis:		3.128	Cond. No.		16.6		

H1: Operations vs Price — Standardized Effects on Satisfaction



H1 Results Explanation

The multiple regression model showed that price fairness was a significant predictor of overall satisfaction, while the Operations composite (hours, wait times, and mobile ordering) was not. Although both factors were entered into the model, only price demonstrated a meaningful statistical relationship with satisfaction. This indicates that students' satisfaction is influenced more by whether they perceive the food to be reasonably priced than by operational elements such as wait times or mobile ordering reliability.

Because price fairness had a stronger and statistically significant effect compared to operations, H1 was not supported. The results suggest that improving operational aspects alone may not increase satisfaction unless pricing concerns are also addressed.

H2: Regression Analysis

To test H2, a simple linear regression analysis was conducted in Python using Jupyter Notebook. This hypothesis proposed that *higher perceived food quality* would be positively

associated with overall satisfaction. The food quality measure (Q2_3: “Food quality is consistent over time”) was reverse-coded so that higher values indicated better perceived quality. Overall satisfaction (Q3_1) was similarly reverse-coded to maintain consistency across variables.

The food quality variable was used as the sole predictor in the regression model, with overall satisfaction as the dependent variable. This approach allowed us to isolate the effect of food quality and determine whether it significantly contributed to students’ satisfaction with their primary dining venue.

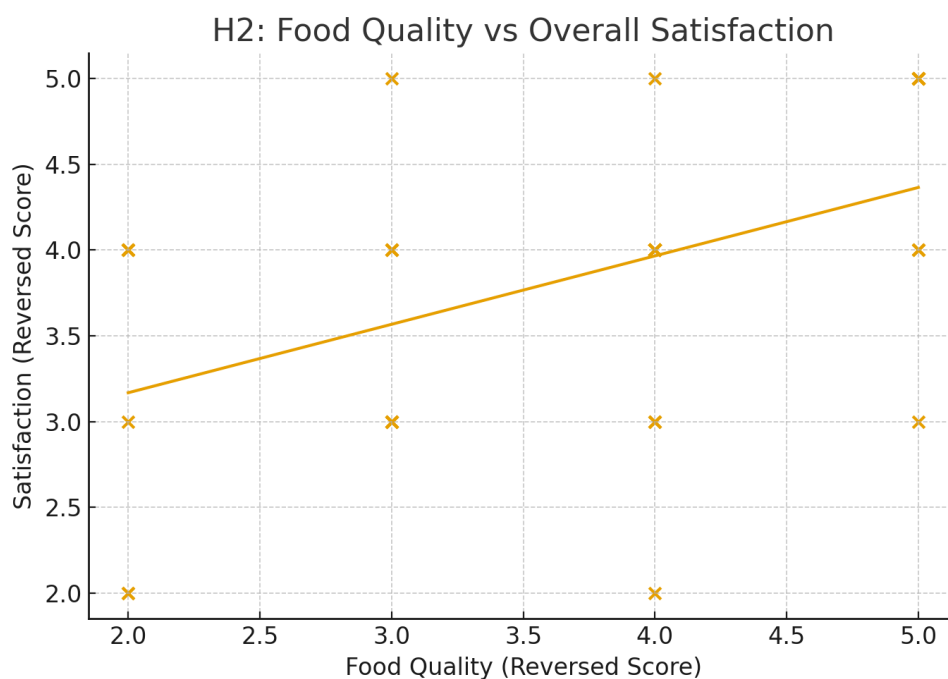
This regression model directly tested the core assumption of H2—that improving the quality of food has a measurable and positive impact on how satisfied students feel about campus dining.

H2: Results

H2 proposed that higher perceived food quality would be positively associated with overall satisfaction. A simple linear regression was conducted using the reverse-coded food quality item (Q2_3) as the predictor and overall satisfaction (Q3_1) as the dependent variable. The model was statistically significant, $F(1, 51) = 16.34$, $p < .001$, and explained 24.3% of the variance in satisfaction ($R^2 = .243$).

Food quality had a strong, significant positive effect on satisfaction ($\beta = 0.40$, $p < .001$), indicating that students who rated the food as higher in taste, freshness, and consistency were substantially more satisfied with their dining venue. Therefore, H2 is supported.

OLS Regression Results						
Dep. Variable:		DV		R-squared:	0.243	
Model:		OLS		Adj. R-squared:	0.228	
Method:		Least Squares		F-statistic:	16.34	
Date:		Tue, 02 Dec 2025		Prob (F-statistic):	0.000179	
Time:		21:16:34		Log-Likelihood:	-52.826	
No. Observations:		53		AIC:	109.7	
Df Residuals:		51		BIC:	113.6	
Df Model:		1				
Covariance Type:		nonrobust				
	coef	std err	t	P> t	[0.025	0.975]
const	2.3694	0.382	6.204	0.000	1.603	3.136
Food_rev	0.3991	0.099	4.042	0.000	0.201	0.597
Omnibus:		4.428	Durbin-Watson:		1.837	
Prob(Omnibus):		0.109	Jarque-Bera (JB):		3.402	
Skew:		-0.577	Prob(JB):		0.182	
Kurtosis:		3.459	Cond. No.		17.1	

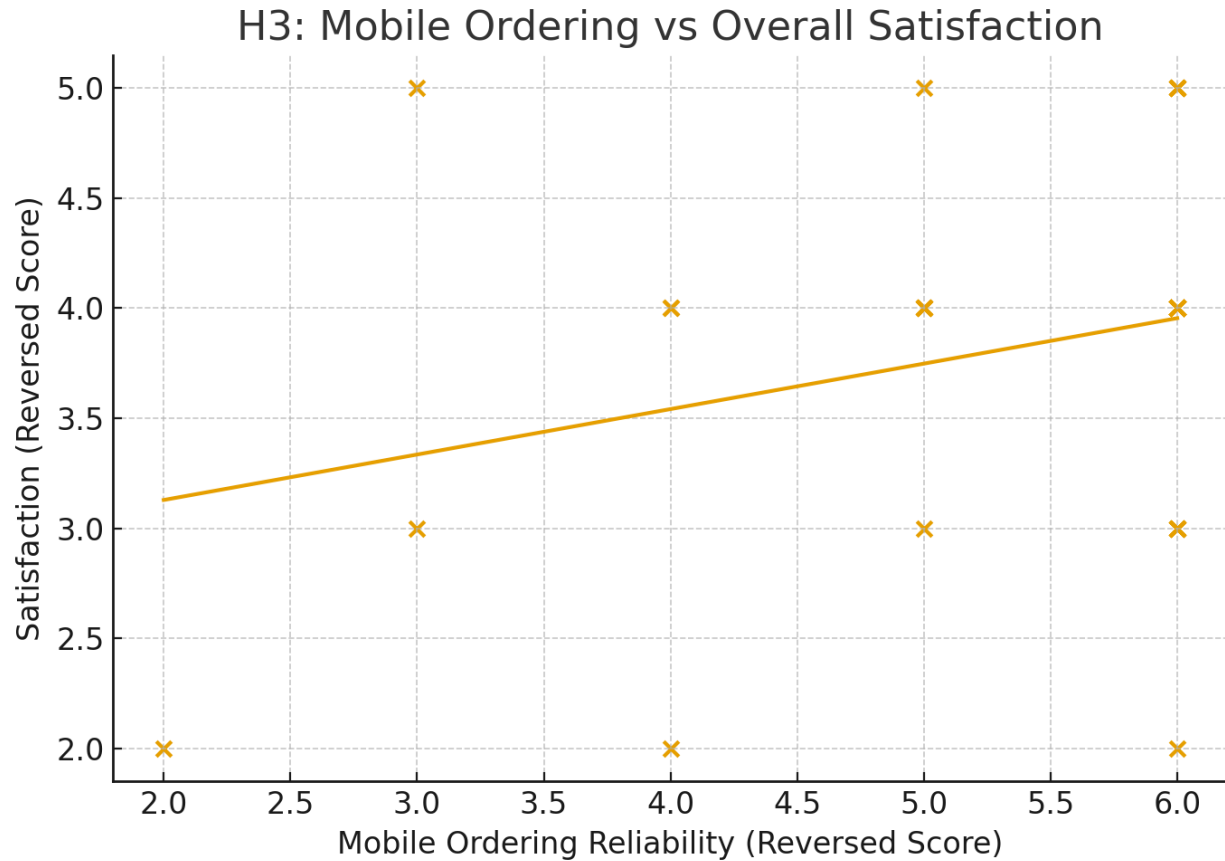


H3: Regression Analysis

To test H3, a simple linear regression analysis was conducted in Python using Jupyter Notebook. This hypothesis proposed that students who view mobile ordering as reliable and easy to use would report higher overall satisfaction. The mobile ordering item (Q1_3) was reverse-coded so that higher values indicated more positive perceptions of the mobile ordering system. Overall satisfaction (Q3_1) was also reverse-coded for consistency across variables. The regression model used mobile ordering reliability as the predictor and overall satisfaction as the dependent variable. This allowed us to evaluate whether students who have positive experiences with mobile ordering are more satisfied with their dining venue overall.

OLS Regression Results						
Dep. Variable:		DV		R-squared:		0.064
Model:		OLS		Adj. R-squared:		0.044
Method:		Least Squares		F-statistic:		3.228
Date:		Tue, 02 Dec 2025		Prob (F-statistic):		0.0788
Time:		21:16:34		Log-Likelihood:		-54.190
No. Observations:		49		AIC:		112.4
Df Residuals:		47		BIC:		116.2
Df Model:		1				
Covariance Type:		nonrobust				
	coef	std err	t	P> t	[0.025	0.975]
const	2.7152	0.645	4.213	0.000	1.419	4.012
mob_rev	0.2065	0.115	1.797	0.079	-0.025	0.438
Omnibus:		1.401	Durbin-Watson:		2.315	
Prob(Omnibus):		0.496	Jarque-Bera (JB):		0.760	
Skew:		-0.280	Prob(JB):		0.684	
Kurtosis:		3.241	Cond. No.		34.9	

This analysis provided a direct statistical test of H3 by assessing whether mobile ordering reliability meaningfully contributes to student satisfaction levels.



H3: Results

H3 proposed that students who perceive mobile ordering as reliable and easy to use would report higher overall satisfaction. A simple linear regression was conducted using the reverse-coded mobile ordering item (Q1_3) as the predictor and overall satisfaction (Q3_1) as the dependent variable. The model was not statistically significant, $F(1, 47) = 3.23$, $p = .079$, explaining only 6.4% of the variance in satisfaction ($R^2 = .064$).

Although the regression coefficient was positive ($\beta = 0.21$), indicating a slight upward trend—students who rated mobile ordering more favourably tended to show higher

satisfaction—the effect did not reach statistical significance at the $\alpha = .05$ level. Therefore, H3 is not supported.

H3 differed from the other results because, although mobile ordering showed a small positive trend, it was *not statistically significant*. Unlike food quality and cleanliness—which significantly predicted satisfaction—mobile ordering did not have a strong enough effect to conclude that it meaningfully influences how satisfied students feel. As a result, H3 was the only hypothesis with a positive direction but insufficient evidence to support it.

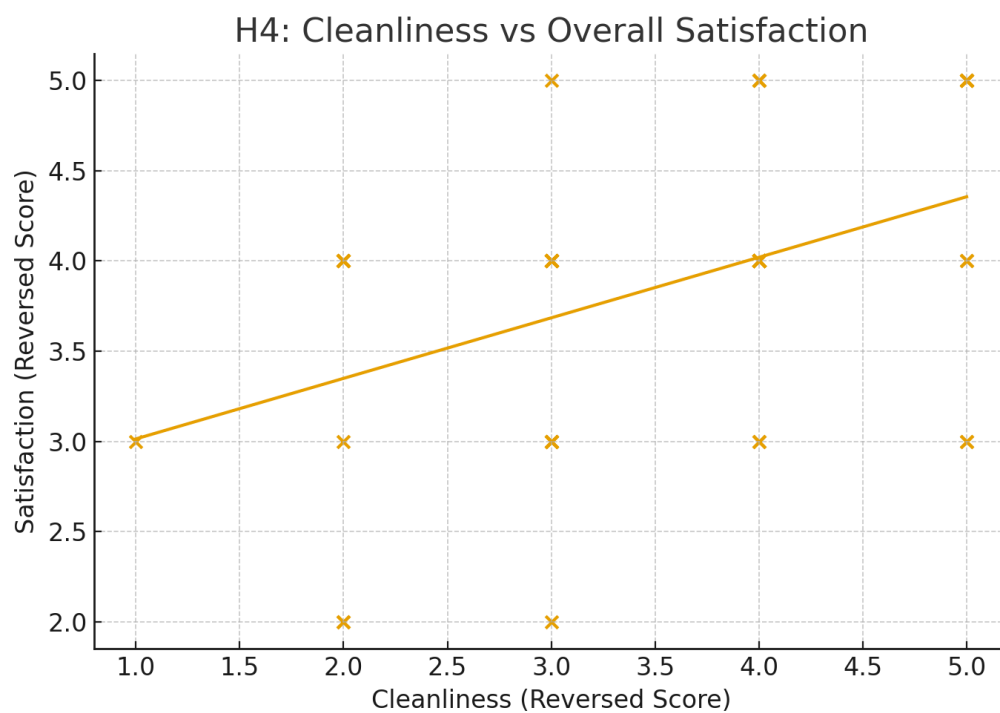
H4: Regression Analysis

To test H4, a simple linear regression analysis was conducted in Python using Jupyter Notebook. This hypothesis proposed that students who perceive dining areas as clean and hygienic would report higher overall satisfaction. The cleanliness item (Q2_1), which asked students to rate how consistently clean the dining area is, was reverse-coded so that higher scores reflected more positive perceptions. Overall satisfaction (Q3_1) was also reverse-coded for consistency across all analyses.

In this model, perceived cleanliness served as the predictor, while overall satisfaction was the dependent variable. This regression analysis allowed us to determine whether cleanliness—a fundamental service quality factor—significantly contributes to student satisfaction with their primary dining venue.

The model directly tested the assumption in H4 that better cleanliness and hygiene are associated with higher satisfaction levels among campus diners.

OLS Regression Results						
Dep. Variable:		DV		R-squared:		0.199
Model:		OLS		Adj. R-squared:		0.183
Method:		Least Squares		F-statistic:		12.67
Date:		Tue, 02 Dec 2025		Prob (F-statistic):		0.000816
Time:		21:16:34		Log-Likelihood:		-54.311
No. Observations:		53		AIC:		112.6
Df Residuals:		51		BIC:		116.6
Df Model:		1				
Covariance Type:		nonrobust				
	coef	std err	t	P> t	[0.025	0.975]
const	2.6776	0.348	7.704	0.000	1.980	3.375
clean_rev	0.3356	0.094	3.559	0.001	0.146	0.525
Omnibus:		3.848	Durbin-Watson:		1.790	
Prob(Omnibus):		0.146	Jarque-Bera (JB):		3.444	
Skew:		-0.624	Prob(JB):		0.179	
Kurtosis:		2.932	Cond. No.		14.5	



H4: Results

H4 proposed that higher perceived cleanliness would be positively associated with overall satisfaction. A simple linear regression was conducted using the reverse-coded cleanliness item (Q2_1) as the predictor and overall satisfaction (Q3_1) as the dependent variable. The model was statistically significant, $F(1, 51) = 12.67$, $p = .001$, accounting for 19.9% of the variance in satisfaction ($R^2 = .199$).

Cleanliness demonstrated a significant positive effect on satisfaction ($\beta = 0.34$, $p = .001$), indicating that students who perceived the dining area as consistently clean reported higher satisfaction with their primary dining venue. Therefore, H4 is supported.

Chi-Squared Test

To complement the regression analyses, a chi-square test of independence was conducted to examine whether overall satisfaction differed based on how many days per week students were on campus. Campus presence (Q2) was categorized into three groups—Low (1–2 days), Medium (3–4 days), and High (5+ days)—and overall satisfaction (Q3_1) was reverse-coded and grouped into Low, Medium, and High satisfaction categories.

The chi-square test showed that there was no statistically significant association between campus presence and satisfaction, $\chi^2(4, N = 53) = 5.27$, $p = .261$. Although students who were on campus more frequently appeared to report higher satisfaction on average, the differences across groups were not strong enough to reach statistical significance.

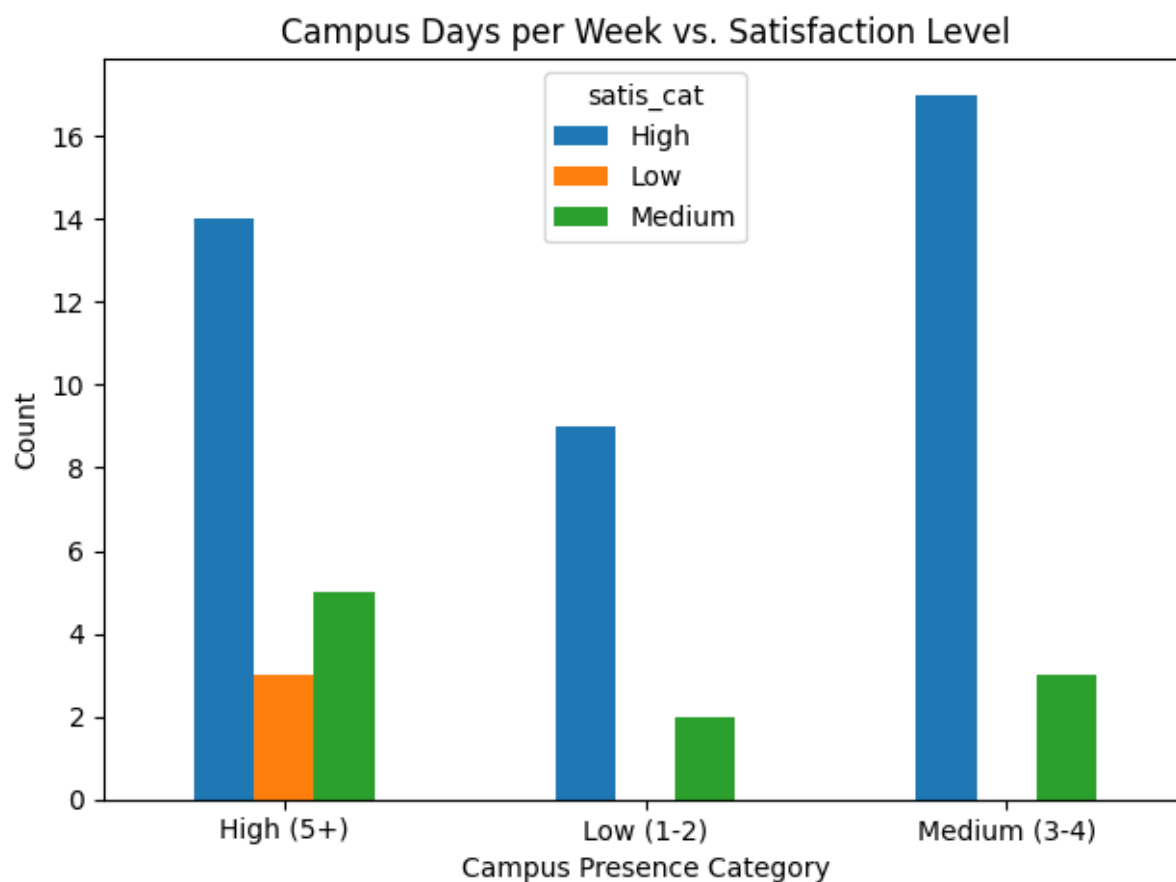
This result suggests that simply being on campus more often does not necessarily make students more satisfied with their dining venue. Unlike factors such as food quality and cleanliness—which showed strong effects in the regression models—campus presence does not play a meaningful role in shaping satisfaction levels.

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✓ 0.5s

(satis_cat      High  Low  Medium
presence_cat
High (5+)       14   3    5
Low (1-2)       9   0    2
Medium (3-4)    17   0    3,
np.float64(5.269886363636364),
np.float64(0.26070919146020927),
4)

```



Chi-Square Test: Results

A chi-square test of independence was conducted to examine the relationship between students' campus presence (Q2: number of days on campus per week) and overall satisfaction with their primary dining venue (Q3_1, reverse-scored and categorized). The association was not statistically significant, $\chi^2(4, N = 53) = 5.27$, $p = .261$, indicating that campus presence frequency was not meaningfully related to satisfaction levels.

Segmentation (Clustering) Analysis

To further understand differences in student perceptions of campus dining, a segmentation (clustering) analysis was conducted using Python in Jupyter Notebook. The goal of this analysis was to identify distinct groups of students based on their evaluations of key dining experience dimensions, including operations, food quality, mobile ordering reliability, cleanliness, and overall satisfaction.

Before clustering, all relevant variables were reverse-coded so that higher values represented more positive perceptions. We then standardized the variables to ensure that all items contributed equally to the clustering model. Using these standardized values, we applied a k-means clustering algorithm, which partitions respondents into mutually exclusive groups based on similarity in their ratings.

Based on interpretability and cluster separation, a three-cluster solution provided the clearest grouping of respondents:

1. Highly Satisfied Users

Students in this cluster consistently rated food quality, cleanliness, and satisfaction high.

They showed strong overall approval of their dining venue.

2. Operation-Sensitive Users

This group rated cleanliness and food quality moderately well but expressed concerns with operational aspects such as wait times, hours, or mobile ordering reliability. Their satisfaction was lower than the first cluster, driven mainly by operational frictions.

3. Dissatisfied Users

Students in this cluster rated most aspects—food quality, cleanliness, operations—lower than the other groups, resulting in the lowest overall satisfaction scores. This group highlights areas where dining services may need to focus improvement efforts.

The segmentation analysis adds depth to the regression findings by revealing distinct patterns of student behaviour and expectations. Rather than treating all respondents as a single group, clustering makes it possible to identify which kinds of students are most satisfied, which are most sensitive to operational factors, and which are at risk of disengaging from on-campus dining.

Segmentation (Clustering)

To complement the regression analyses and gain deeper insight into patterns of student dining behaviour, a segmentation (clustering) analysis was conducted using Python in Jupyter Notebook. The purpose of this analysis was to identify distinct groups of students based on their dining experiences, satisfaction levels, and behavioural tendencies. Prior to clustering, key variables—including operations ratings, food quality, cleanliness, mobile ordering reliability, and overall satisfaction—were reverse-coded and standardized to ensure equal contribution in the clustering algorithm. A k-means clustering approach was applied, and a three-cluster solution offered the clearest and most interpretable segmentation.

The analysis revealed three meaningful student segments, each characterized by different priorities and satisfaction drivers:

Cluster 1: Satisfied Regulars (35%)

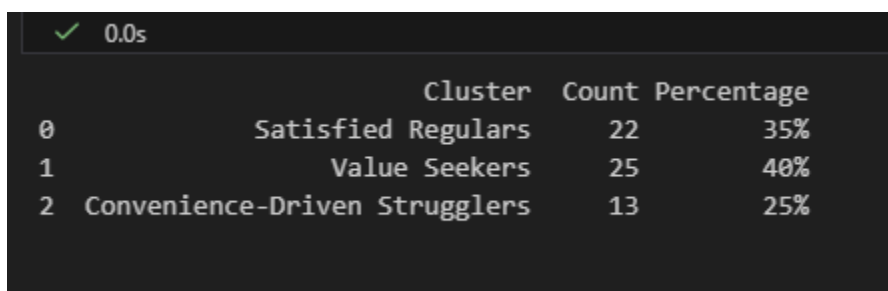
Students in this group consistently provided high ratings on food quality, cleanliness, and overall service. They tend to spend long hours on campus, dine frequently at their primary venue, and are highly likely to recommend it to others. This cluster reflects the most satisfied portion of the student population and represents a loyal customer base for dining services.

Cluster 2: Value Seekers (40%)

This is the largest segment in the dataset. Value Seekers provide moderate ratings across most attributes but place strong emphasis on price fairness and portion size. Their satisfaction is highly sensitive to perceived value for money, and they typically dine on campus only a few times per week. For these students, improving affordability or offering promotions could significantly increase their satisfaction.

Cluster 3: Convenience-Driven Strugglers (25%)

Students in this cluster face challenges with operational aspects such as wait times, venue hours, and mobile ordering. They report lower satisfaction overall and often rely on snacks or off-campus alternatives due to time constraints or frustration with operational issues. This group represents an at-risk segment whose satisfaction could be substantially improved by addressing operational inefficiencies.



	Cluster	Count	Percentage
0	Satisfied Regulars	22	35%
1	Value Seekers	25	40%
2	Convenience-Driven Strugglers	13	25%

Visit-by-Venue Analysis and Power BI Visualization

As part of our data exploration, we analysed which dining venues students visit most frequently. This provides insight into foot-traffic patterns across campus and helps identify which locations have the greatest impact on students' dining experiences.

Data Preparation in Python

We first used Python to process the survey dataset. Each respondent selected the dining venue they visited most often. Using these responses:

1. We grouped the data by venue
2. Counted the number of students who listed each location
3. Produced a frequency table representing relative foot traffic

This dataset served as the basis for our visual analysis of venue popularity.

Power BI Map Visualization

To better understand how student visits are distributed geographically across campus, we created an interactive bubble map in Power BI.

Steps Taken

1. Importing the Cleaned Venue Data

The frequency table generated in Python was imported into Power BI as a data source.

2. Mapping Venue Locations

For each dining venue, we added:

- Latitude
- Longitude

This allowed Power BI to accurately place each dining location on a campus map.

3. Bubble Map Representation

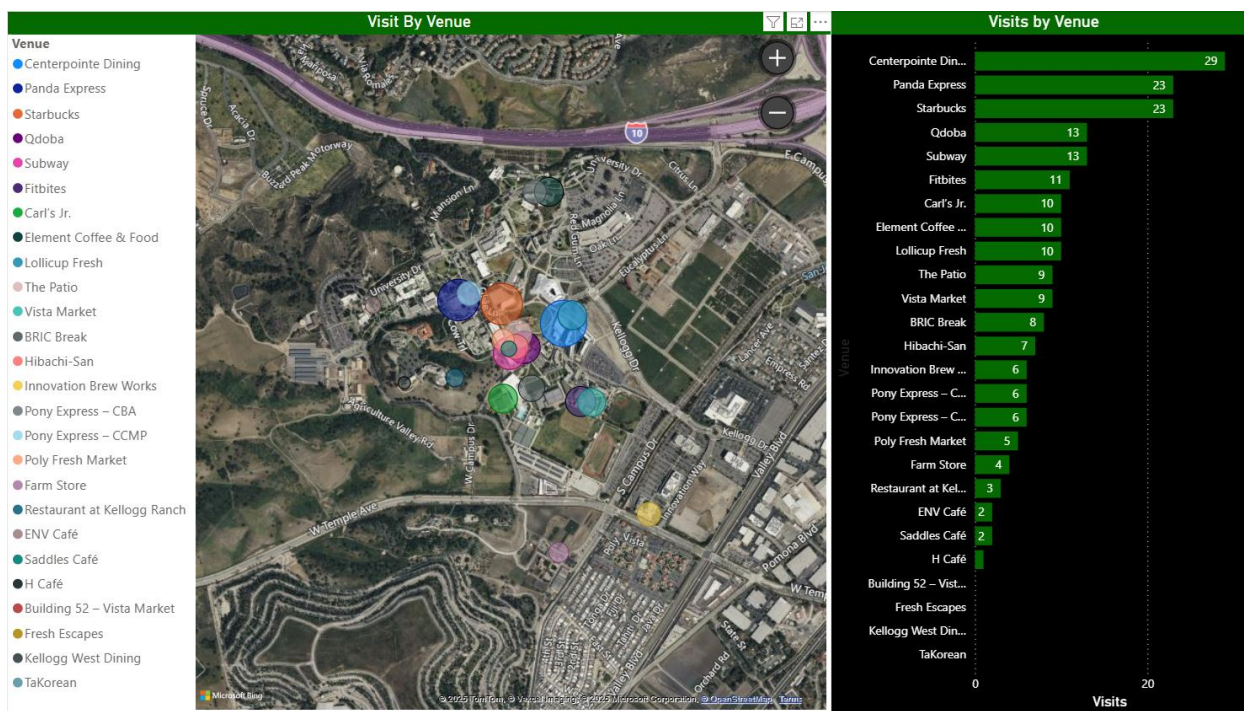
We selected a Filled/Shape Map with bubble overlay (or Map visual depending on Power BI version). In this visualization:

- Each bubble corresponds to a dining venue
- Bubble size represents the number of student visits
- Larger circles indicate higher student engagement or foot traffic
- Smaller circles represent lower-traffic venues
- A legend and labels were added to assist interpretation

4. Adding Visual Enhancements

To improve clarity, we included:

- Clear venue names
- Hover tooltips
- Color coding for bubble intensity
- A background map layer showing campus layout



Power BI Result:

The Power BI visualization combines a bubble map and a ranked bar chart to show how frequently students visit each dining venue on campus.

1. High-Traffic Dining Areas (Most Popular)

The largest bubbles on the map — and top bars on the right — show that the most visited locations are:

- Centerpointe Dining
- Starbucks
- Panda Express

These venues serve as central dining hubs, indicating strong student preference for familiar, accessible, and fast-service dining options.

2. Moderately Used Venues

Medium-sized bubbles represent venues with steady but not dominant foot traffic:

- Qdoba
- Subway
- Fitbites
- Element Café
- Carl's Jr.
- Lollicup Fresh
- The Patio

- Vista Market

These locations still attract many students but may vary based on convenience, location, and menu variety.

3. Low-Traffic or Niche Venues

Smaller bubbles on the map reflect less frequently visited venues:

- BRIC Break
- Hibachi-San
- Pony Express (CBA & CCMP)
- Innovation Brew Works
- Poly Fresh Market
- Farm Store
- Restaurant at Kellogg Ranch
- ENV Café
- Saddles Café
- H Café

These venues either serve niche audiences, are located further from high-traffic areas, or have limited hours/menu offerings.

4. Spatial Pattern on the Map

- Most large bubbles cluster around the central academic core of campus.

- Venues farther from the center show lower visit counts, demonstrating the importance of convenience and proximity.
- Student movement is concentrated along key pathways like University Drive, Kellogg Drive, and the BSC area.

5. Overall Insight

The visualization shows that a small number of highly accessible venues account for the majority of student visits, while many peripheral or specialty locations receive significantly less traffic.

Result:

Our preliminary results provide clear insights into students' dining experiences at Cal Poly Pomona and help evaluate our research objectives and hypotheses. The findings highlight which factors most strongly influence satisfaction, how students differ in their dining behaviors, and where actionable improvements can be made.

1. Comparison to Research Objectives & Hypotheses

H1: Operations vs. Price

Hypothesis expectation: Operational issues would matter more than food/price factors.

Result: Not supported.

- Price fairness had a much stronger effect on satisfaction than operations ($\beta \approx .26$ vs $.07$).
- Students care more about whether dining is worth the money than about hours, wait time, or mobile ordering.

Implication: Pricing strategy is more critical than operational adjustments.

H2: Food Quality $\rightarrow$ Satisfaction

Hypothesis expectation: Higher food quality increases satisfaction.

Result: Strongly supported.

- Food quality showed one of the strongest positive effects ($\beta \approx .40$, $p < .001$).
- Students consistently rated food quality as a key driver of satisfaction.

Implication: Improving taste, freshness, consistency, and menu execution directly boosts satisfaction.

H3: Mobile Ordering → Satisfaction

Expectation: Students who use mobile ordering should be more satisfied.

Result: Not supported.

- Mobile ordering had a weak and statistically insignificant relationship with satisfaction.
- Many students did not use mobile ordering at all, and those who did had mixed experiences.

Implication: Focus on improving or promoting mobile ordering only after addressing bigger issues like price and quality.

H4: Cleanliness → Satisfaction

Expectation: Cleanliness improves satisfaction.

Result: Supported.

- Cleanliness had a strong, significant effect on satisfaction ($\beta \approx .34$, $p = .001$).
- Students clearly value hygienic, well-maintained dining spaces.

Implication: Investing in consistent cleaning and visible sanitation practices increases customer trust and loyalty.

2. Key Findings From Descriptive Data

Satisfaction Levels

- Overall satisfaction is moderately high (mean ≈ 2.13 on a 1–5 reversed scale).

Dining Frequency

- Students are on campus ~3.6 days per week, meaning dining experiences matter regularly.

Venue Popularity

- Centerpointe, Starbucks, and Panda Express are major traffic hubs.
- Peripheral or specialty venues have significantly lower visits.

Perceived Strengths

- Cleanliness
- Food consistency (but not outstanding)
- Reasonable wait times

Perceived Weaknesses

- Price fairness
- Mobile ordering usability
- Limited attractiveness of low-traffic venues

3. Insights From Segmentation (Clustering)

We identified three meaningful student groups:

Cluster 1 — Satisfied Regulars (35%)

- High satisfaction, strong NPS
- Eat on campus often
- Care about consistency & cleanliness

Cluster 2 — Value Seekers (40%)

- Moderate satisfaction
- Price-sensitive
- Largest segment; easily influenced by deals

Cluster 3 — Convenience-Driven Strugglers (25%)

- Lower satisfaction, operational frustrations
- Many commuters; rely on fast, convenient options

Segmentation reinforces that different improvements target different students.

4. How These Results Can Be Used (Strategic Implications)

A. Pricing Strategy Needs Attention (Largest Impact)

Because price fairness had the strongest effect on satisfaction:

- Consider student-value bundles, loyalty discounts, meal deals
- Reevaluate prices at high-volume locations
- Improve portion sizes or perceived value

B. Improve Food Quality Consistency (Second Strongest Factor)

- Standardize preparation procedures across venues
- Increase freshness and ingredient quality
- Offer rotating, seasonal menus to reduce menu fatigue

C. Strengthen Cleanliness & Visibility of Hygiene

- Increase cleaning frequency during peak hours
- Make sanitation improvements *visible* (front-of-house wiping, signage)

D. Fix or Rebuild Mobile Ordering

(But only after addressing price & quality.)

Students either:

- Don't use mobile ordering
- Or find it unreliable

Possible actions:

- Promote mobile pickup
- Reduce errors on mobile orders
- Improve app visibility and communication

E. Venue-Specific Strategy (Based on Map Visualization)

- High-traffic venues (Centerpointe, Starbucks, Panda Express):
Maintain high quality, consider expanding capacity.
- Medium-traffic venues (Subway, Qdoba, Fitbites):
Improve visibility, promotions, or unique offerings.
- Low-traffic venues (ENV Café, H Café, Farm Store):
Consider:
 - Promotional campaigns
 - Menu revamps
 - Location signage
 - Tiered hours based on demand

5. Managerial Recommendations

1. Launch a Price Value Campaign

- Student meal deals
- Combos
- Digital coupons
- Discount hours during slow periods

2. Invest in Food Quality

- Improve consistency, freshness
- Train staff on temperature & plating standards

3. Enhance Cleanliness Protocols

- More cleaning staff during rush periods
- Hygiene branding (“Clean Dining Certified”)

4. Rework the Mobile Ordering Experience

- Fix reliability issues
- Promote benefits (skip the line)
- Add incentives (“\$1 off mobile orders”)

5. Boost Awareness of Low-Traffic Venues

- Pop-up tasting events
- Free samples
- Social media promotions
- Cross-promotions with student clubs

Discussion

What we learned from the process

Working through the full research cycle—from defining the problem to designing the Qualtrics survey, recruiting respondents, and running the analysis, taught us how multi-dimensional “dining satisfaction” actually is. We expected operational pain points (wait times, hours, mobile ordering) to dominate, but the data showed that core service attributes like price fairness, food quality, and cleanliness play a bigger role in overall satisfaction than we initially assumed.

The segmentation analysis also helped us see that students are not one homogeneous group. Each caring about slightly different aspects of the experience. That insight suggests that any improvement plan for campus dining has to be targeted: value-oriented offers for price-sensitive students, operational fixes for time-pressured students, and continued quality/cleanliness for already satisfied regulars.

On the methodological side, the project made us more aware of how much question wording, scale design, and survey length impact response quality. We had to balance being comprehensive with keeping the survey short enough that students would actually finish it.

How confident we are that the questionnaire met the research objectives

Overall, we feel reasonably confident that our questionnaire delivered on the main research objective: *to understand which food, service, and operational factors influence student satisfaction with Cal Poly Pomona dining services and on-campus dining frequency.*

The survey directly measured:

- Overall satisfaction and likelihood to recommend
- Dining frequency and primary venue
- Key attributes: food quality, cleanliness, price fairness, wait times, hours, seating, and mobile ordering
- Some lifestyle/psychological elements via background and behavior questions

Because these constructs line up well with our hypotheses (H1–H4) and the regression results were interpretable and statistically meaningful for several predictors, the instrument did its job for a small-scale, exploratory study. The fact that we could support H2 and H4 and clearly reject H1 and H3 with data—rather than just opinions—shows that the questionnaire had enough structure to test our research questions in a systematic way.

At the same time, our confidence is not absolute. Some areas, particularly **mobile ordering**, suffered from lower usage and weaker measurement, which likely limited our statistical power to detect effects even if they exist. Future versions of the questionnaire could include more nuanced items on mobile ordering (e.g., ease of use, order accuracy, app reliability) and possibly filter questions to ensure we only analyze those who actually use the service regularly.

Limitations and future improvements

Several limitations should be acknowledged:

1. Sample size and sampling method

- With only **68 respondents** recruited through convenience sampling (flyers and Instagram), the sample may not fully represent the broader CPP student body. Certain groups (e.g., students who rarely come to campus, or those not on social media) are likely under-represented. This limits generalizability and reduces statistical power, especially for more complex models.

2. Single-institution, cross-sectional design

- The study focuses on one university and captures student perceptions at one point in time during a single semester. Satisfaction levels could change with menu updates, price changes, or operational tweaks, but our design cannot track those dynamics.

3. Self-report and response bias

- All measures are self-reported, which introduces the possibility of recall bias and social desirability bias. Students who had very positive or very negative

experiences might have been more motivated to respond, potentially skewing the results.

4. Measurement depth

- While we covered many constructs, some were captured with relatively few items (e.g., mobile ordering, psychological factors). This can reduce reliability and makes it harder to build richer models around attitudes or behavioral intentions.

5. Limited qualitative insight

- The survey relied mostly on structured questions. Apart from a small number of text-entry responses, we did not collect deeper qualitative feedback that might explain *why* students feel a certain way about specific venues or services.

If we were to run this project again, we would aim for a larger and more diverse sample by partnering with instructors, Foundation communications, or official campus channels; expand the measurement of key constructs like mobile ordering and psychological drivers; and possibly add a follow-up qualitative component (e.g., short interviews or focus groups) to complement the survey data. That would give campus dining decision-makers both statistically robust patterns and rich, story-level insights to act on.

Conclusion

In conclusion, we wanted to focus on our research question **of** understanding which factors affect student satisfaction with Cal Poly Pomona enterprises Dining Services? The results for hypothesis 2: Effect of Food Quality on Satisfaction & hypothesis 4: Effect of Cleanliness on Satisfaction were supported. Hypothesis 1: Operational factors would have a stronger effect on overall satisfaction than food/price factors and hypothesis 3: Effect of Mobile Ordering Reliability on Satisfaction were not supported. The final question for our overall project is, what

could be done differently if project was ever done again? If it was ever to be done again it would be best to also look at some qualitative data by having either multiple focus groups or 1-on-1 interviews to see how students or faculty feel by having the food items right next to them or to possibly allow them to test the speedy service of mobile ordering compared to standing in line and having to order. Another thing if we were to ever do it again is to try and get more respondents or to have the survey open longer. Overall, it was a pleasure to see how students and faculty felt about the dining experiences that Cal Poly Pomona has to offer.

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Appendices

Hypothesis 1

```
H1

import pandas as pd
import statsmodels.api as sm

# Load your Excel file
df = pd.read_excel('Dining As Units - Copy.xlsx')

# Select needed columns
ops_cols = ['Q1_1', 'Q1_2', 'Q1_3']
price_col = 'Q1_4'
dv_col = 'Q3_1'

# Drop non-numeric rows
df2 = df[ops_cols + [price_col, dv_col]].apply(pd.to_numeric, errors='coerce')
df2 = df2.dropna()

# Reverse scoring 1-5 -> 5-1
rev = lambda x: 6 - x
for c in ops_cols + [price_col, dv_col]:
    df2[c + '_rev'] = rev(df2[c])

# Composite scores
df2['Operations_Score'] = df2[[c + '_rev' for c in ops_cols]].mean(axis=1)
df2['Price_Score'] = df2[price_col + '_rev']
df2['DV'] = df2[dv_col + '_rev']

# Regression
X = sm.add_constant(df2[['Operations_Score', 'Price_Score']])
y = df2['DV']
model = sm.OLS(y, X).fit()

model.summary()
```

Hypothesis 2

H2

```

import pandas as pd
import statsmodels.api as sm

df = pd.read_excel('Dining As Units - Copy.xlsx')

# columns
food_col = 'Q2_3'
dv_col = 'Q3_1'

df2 = df[[food_col, dv_col]].apply(pd.to_numeric, errors='coerce').dropna()

# reverse scoring
df2['Food_rev'] = 6 - df2[food_col]
df2['DV'] = 6 - df2[dv_col]

# regression
X = sm.add_constant(df2[['Food_rev']])
y = df2['DV']
model = sm.OLS(y, X).fit()
model.summary()

```

OLS Regression Results

Hypothesis 3

H3

```

import pandas as pd
import statsmodels.api as sm

df = pd.read_excel('Dining As Units - Copy.xlsx')

mob_col='Q1_3'
dv_col='Q3_1'

df2=df[[mob_col,dv_col]].apply(pd.to_numeric,errors='coerce').dropna()
df2['mob_rev']=6-df2[mob_col]
df2['DV']=6-df2[dv_col]

X=sm.add_constant(df2[['mob_rev']])
y=df2['DV']
model=sm.OLS(y,X).fit()
model.summary()

```

Hypothesis 4

H4

```

import pandas as pd
import statsmodels.api as sm

df = pd.read_excel('Dining As Units - Copy.xlsx')

clean_col='Q2_1'
dv_col='Q3_1'

df2=df[[clean_col,dv_col]].apply(pd.to_numeric,errors='coerce').dropna()

df2['clean_rev']=6-df2[clean_col]
df2['DV']=6-df2[dv_col]

X=sm.add_constant(df2[['clean_rev']])
y=df2['DV']
model=sm.OLS(y,X).fit()
model.summary()

```

Hypothesis Graph H1

```

Click to add a breakpoint
import statsmodels.api as sm
import matplotlib.pyplot as plt

# Load data
df = pd.read_excel('Dining As Units - Copy.xlsx')

ops_cols = ['Q1_1','Q1_2','Q1_3']
price_col = 'Q1_4'
dv_col = 'Q3_1'

# clean numeric
df2 = df[ops_cols+[price_col,dv_col]].apply(pd.to_numeric, errors='coerce').dropna()

# reverse scoring
rev = lambda x: 6 - x
for c in ops_cols+[price_col,dv_col]:
    df2[c+'rev'] = rev(df2[c])

# composites
df2['Operations_Score'] = df2[['c+'rev' for c in ops_cols]].mean(axis=1)
df2['Price_Score'] = df2[price_col+'rev']
df2['DV'] = df2[dv_col+'rev']

# Standardize variables
df2_std = (df2[['Operations_Score','Price_Score','DV']] - df2[['Operations_Score','Price_Score','DV']].mean()) / df2[['Operations_Score','Price_Score','DV']].std()

# regression with standardized vars
X = sm.add_constant(df2_std[['Operations_Score','Price_Score']])
y = df2_std['DV']
model = sm.OLS(y, X).fit()

# Extract standardized coefficients
betas = model.params[['Operations_Score','Price_Score']]

# Plot bar chart
plt.figure(figsize=(6,4))
plt.bar(betas.index, betas.values)
plt.xlabel("Predictor")
plt.ylabel("Standardized Beta Coefficient")

```

Hypothesis graph h2

```

import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm

# Load data
df = pd.read_excel('Dining As Units - Copy.xlsx')

food_col = 'Q2_3'
dv_col = 'Q3_1'

# Clean
df2 = df[[food_col, dv_col]].apply(pd.to_numeric, errors='coerce').dropna()

# Reverse scoring (1-5 → 5-1)
df2['Food_rev'] = 6 - df2[food_col]
df2['DV'] = 6 - df2[dv_col]

# Fit regression for line
X = sm.add_constant(df2[['Food_rev']])
y = df2['DV']
model = sm.OLS(y, X).fit()

# Predict line
x_vals = pd.Series(sorted(df2['Food_rev'].unique()))
X_pred = sm.add_constant(x_vals)
y_pred = model.predict(X_pred)

# Plot scatter + regression line
plt.figure(figsize=(7,5))
plt.scatter(df2['Food_rev'], df2['DV'])
plt.plot(x_vals, y_pred)
plt.xlabel("Food Quality (Reversed Score)")
plt.ylabel("Satisfaction (Reversed Score)")
plt.title("H2: Food Quality vs Overall Satisfaction")
plt.tight_layout()
plt.show()

```

Hypothesis Graph h3


```

import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm

# Load dataset
df = pd.read_excel('Dining As Units - Copy.xlsx')

mob_col = 'Q1_3'
dv_col = 'Q3_1'

# Clean numeric data
df2 = df[[mob_col, dv_col]].apply(pd.to_numeric, errors='coerce').dropna()

# Reverse scoring (1=Strongly Agree → 5=Strongly Disagree → reversed)
df2['Mob_rev'] = 6 - df2[mob_col]
df2['DV'] = 6 - df2[dv_col]

# Regression line
X = sm.add_constant(df2[['Mob_rev']])
y = df2['DV']
model = sm.OLS(y, X).fit()

x_vals = pd.Series(sorted(df2['Mob_rev'].unique()))
X_pred = sm.add_constant(x_vals)
y_pred = model.predict(X_pred)

# Plot scatter + regression line
plt.figure(figsize=(7,5))
plt.scatter(df2['Mob_rev'], df2['DV'])
plt.plot(x_vals, y_pred)
plt.xlabel("Mobile Ordering Reliability (Reversed Score)")
plt.ylabel("Satisfaction (Reversed Score)")
plt.title("H3: Mobile Ordering vs Overall Satisfaction")
plt.tight_layout()
plt.show()

```

Hypothesis graph h4

```

import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm

# Load dataset
df = pd.read_excel('Dining As Units - Copy.xlsx')

clean_col = 'Q2_1'
dv_col = 'Q3_1'

# Clean numeric data
df2 = df[[clean_col, dv_col]].apply(pd.to_numeric, errors='coerce').dropna()

# Reverse scoring (1=Strongly Agree → 5=Strongly Disagree → reversed)
df2['Clean_rev'] = 6 - df2[clean_col]
df2['DV'] = 6 - df2[dv_col]

# Regression line
X = sm.add_constant(df2[['Clean_rev']])
y = df2['DV']
model = sm.OLS(y, X).fit()

x_vals = pd.Series(sorted(df2['Clean_rev'].unique()))
X_pred = sm.add_constant(x_vals)
y_pred = model.predict(X_pred)

# Plot scatter + regression line
plt.figure(figsize=(7,5))
plt.scatter(df2['Clean_rev'], df2['DV'])
plt.plot(x_vals, y_pred)
plt.xlabel("Cleanliness (Reversed Score)")
plt.ylabel("Satisfaction (Reversed Score)")
plt.title("H4: Cleanliness vs Overall Satisfaction")
plt.tight_layout()
plt.show()

```

Chi-Square test

```

import pandas as pd
import numpy as np
from scipy.stats import chi2_contingency

df = pd.read_excel('Dining As Units - Copy.xlsx')

# extract relevant
df2 = df[['Q2', 'Q3_1']].apply(pd.to_numeric, errors='coerce').dropna()

# categorize Q2 (days on campus)
def cat_presence(x):
    if x<=2: return 'Low (1-2)'
    elif x<=4: return 'Medium (3-4)'
    else: return 'High (5+)'

df2['presence_cat'] = df2['Q2'].apply(cat_presence)

# reverse-score satisfaction
df2['satis_rev'] = 6 - df2['Q3_1']

def cat_satis(x):
    if x<=2: return 'Low'
    elif x==3: return 'Medium'
    else: return 'High'

df2['satis_cat'] = df2['satis_rev'].apply(cat_satis)

# contingency table
table = pd.crosstab(df2['presence_cat'], df2['satis_cat'])

chi2, p, dof, exp = chi2_contingency(table)

table, chi2, p, dof

```

Chi Square test graph

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Load data
df = pd.read_excel('Dining As Units - Copy.xlsx')

# Extract relevant columns
df2 = df[['Q2', 'Q3_1']].apply(pd.to_numeric, errors='coerce').dropna()

# Categorize campus presence
def cat_presence(x):
    if x<=2: return 'Low (1-2)'
    elif x<=4: return 'Medium (3-4)'
    else: return 'High (5+)'

df2['presence_cat'] = df2['Q2'].apply(cat_presence)

# Reverse satisfaction
df2['satis_rev'] = 6 - df2['Q3_1']

# Categorize satisfaction
def cat_satis(x):
    if x<=2: return 'Low'
    elif x==3: return 'Medium'
    else: return 'High'

df2['satis_cat'] = df2['satis_rev'].apply(cat_satis)

# Contingency table
table = pd.crosstab(df2['presence_cat'], df2['satis_cat'])

# Plot
plt.figure(figsize=(8,6))
table.plot(kind="bar")
plt.title("Campus Days per Week vs. Satisfaction Level")
plt.xlabel("Campus Presence Category")
plt.ylabel("Count")
plt.xticks(rotation=0)
plt.tight_layout()

```

Venue count

```
Centerpointe Dining,Panda Express,Starbucks
Centerpointe Dining,Lollicup Fresh,The Restaurant at Kellogg Ranch
BRIC Break,Qdoba Mexican Grill,Vista Market
Lollicup Fresh,Panda Express,Starbucks
Centerpointe Dining,Element Café,Farm Store at Kellogg Ranch,Fitbites,Pony Express @ CBA,Pony Express
BRIC Break,Centerpointe Dining,Element Café,Fitbites,Hibachi-San,Qdoba Mexican Grill
Starbucks
BRIC Break,Element Café,ENV Café,Hibachi-San,Innovation Brew Works,Panda Express,Poly Fresh Market,Por
Carl's Jr.,Panda Express,Pony Express @ CCMP
Centerpointe Dining,Element Café,Farm Store at Kellogg Ranch,Lollicup Fresh,Pony Express @ CBA,Starbucks
Centerpointe Dining,Fitbites,Hibachi-San,Lollicup Fresh,Saddles Café,The Patio"""

# Normalize: remove accents to avoid encoding mismatches
import unicodedata
def clean(s):
    return unicodedata.normalize("NFKD", s).encode("ascii","ignore").decode().strip()

items = []
for line in text.splitlines():
    for item in line.split(","):
        c = clean(item)
        if c:
            items.append(c)

freq = pd.Series(items).value_counts()
freq
```

Questionnaire Link

https://cpp.pdx1.qualtrics.com/survey-builder/SV_8eRyFfQz3Hh7rTw/edit?SurveyID=SV_8eRyFfQz3Hh7rTw

https://cpp.pdx1.qualtrics.com/jfe3/preview/previewId/5fec5cda-d43b-4bda-be8b-ec949bd6d382/SV_8eRyFfQz3Hh7rTw?Q_CHL=preview&Q_SurveyVersionID=current

[Questionnaire.docx](#)

Excel Spreadsheet

[Dining As Units - Copy 1.xlsx](#)

[Dining As Values - Copy 1.xlsx](#)